Jialuo Du

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Research Interests

Low-power wide-area IoT, wireless physical-layer systems, and machine learning for wireless networks. My work combines physical signal models with learning to improve weak-signal detection, demodulation, and cooperation among LoRa gateways.

Professional Appointments

Michigan State University, *Research Associate (Postdoctoral)* *Aug 2025 – Present*

- Department of Computer Science and Engineering. Supervisor: Prof. Zhichao Cao

Education

Tsinghua University, *Ph.D., School of Software* *Sept 2020 – July 2025*

- Advisor: Prof. Yunhao Liu
- Dissertation: Research on LoRa Weak-Signal Communication Based on Probabilistic Modeling and Deep Learning

Tsinghua University, *B.Eng., Computer Science and Technology* *Sept 2016 – June 2020*

Publications

Conference Papers

DeRa: Deep-Range Low-Power Wide Area IoT via Ultra-sensitive Signal Detection
Jialuo Du, Jingkai Lin, Zhichao Cao
ACM MobiCom 2026, pp. 599–619. DOI

Kairos: Soil-Aware Transmission Control for Long-Term Agricultural Cross-Soil LoRa
Jingkai Lin, Yidong Ren, **Jialuo Du**, Yimeng Liu, Yawen Wang, Younsuk Dong, Tianxing Li, Zhichao Cao
IEEE ICNP 2026. PDF

LoRaSeek: Boosting Denoising Ability in Neural-enhanced LoRa Decoder via Hierarchical Feature Extraction
Khang Nguyen, Yidong Ren, **Jialuo Du**, Jingkai Lin, Maolin Gan, Shigang Chen, Mi Zhang, Chunyi Peng, Zhichao Cao
ACM MobiCom 2025. DOI

LoRaTrimmer: Optimal Energy Condensation with Chirp Trimming for LoRa Weak Signal Decoding
Jialuo Du, Yunhao Liu, Yidong Ren, Li Liu, Zhichao Cao
ACM MobiCom 2024. DOI

Demeter: Reliable Cross-soil LPWAN with Low-cost Signal Polarization Alignment
Yidong Ren, Wei Sun, **Jialuo Du**, Huaili Zeng, Younsuk Dong, Mi Zhang, Shigang Chen, Yunhao Liu, Tianxing Li, Zhichao Cao
ACM MobiCom 2024. DOI

SRLoRa: Neural-enhanced LoRa Weak Signal Decoding with Multi-gateway Super Resolution
Jialuo Du, Yidong Ren, Zhui Zhu, Chenning Li, Zhichao Cao, Qiang Ma, Yunhao Liu
ACM MobiHoc 2023. DOI

Prism: High-throughput LoRa Backscatter with Non-linear Chirps
Yidong Ren, Puyu Cai, Jinyan Jiang, **Jialuo Du**, Zhichao Cao
IEEE INFOCOM 2023. DOI

SRPeek: Super Resolution Enabled Screen Peeking via COTS Smartphone
Jialuo Du, Chenning Li, Zhenge Guo, Zhichao Cao
IEEE ICPADS 2021. DOI

EyeLoc: Smartphone Vision-enabled Plug-n-play Indoor Localization in Large Shopping Malls
Zhichao Cao, **Jialuo Du**, Manni Liu, Qing Zhou
IEEE DySPAN 2019.

Journal Articles

Umami-MRNN: Deep Learning-based Prediction of Umami Peptide Using RNN and MLP
Lulu Qi, **Jialuo Du**, Yue Sun, Yongzhao Xiong, Xinyao Zhao, Daodong Pan, Yueru Zhi, Yali Dang, Xinchang Gao
Food Chemistry, 2023. DOI

EyeLoc: Smartphone Vision-enabled Plug-n-play Indoor Localization in Large Shopping Malls
Manni Liu, **Jialuo Du**, Qing Zhou, Zhichao Cao, Yunhao Liu
IEEE Internet of Things Journal, 2020. DOI

Workshop and Forum Papers

Scaling LoRa to Massive IoT: A Physical Layer Approach
Jialuo Du
ACM MobiSys 2026 Rising Stars Forum (MobiSys Companion '26), pp. 150–151. DOI

NELoRa-Bench: A Benchmark for Neural-enhanced LoRa Demodulation
Jialuo Du, Yidong Ren, Mi Zhang, Yunhao Liu, Zhichao Cao
ICLR 2023 Workshop on Machine Learning for IoT. arXiv:2305.01573

Preprints

AIoT-based Continuous, Contextualized, and Explainable Driving Assessment for Older Adults
Yimeng Liu, Fangwei Zhang, Maolin Gan, **Jialuo Du**, Jingkai Lin, Yawen Wang, Fei Sun, Honglei Chen, Linda Hill, Ruofeng Liu, Tianxing Li, Zhichao Cao
arXiv:2603.00691, 2026. arXiv

Research Highlights

Ultra-sensitive packet detection (DeRa, MobiCom 2026, first author)

- Identified packet detection as a bottleneck in weak-signal LoRa reception and modeled carrier-frequency-offset-induced phase evolution to enable coherent preamble accumulation.
- Reported a 4.3–4.8 dB lower SNR threshold than LoRaTrimmer for at least 90% packet detection in the evaluated settings, using commodity LoRa transmitters.

Optimal weak-signal demodulation (LoRaTrimmer, MobiCom 2024, first author)

- Modeled intra-symbol frequency and phase discontinuities in commodity LoRa signals and derived a decoder that is optimal under the paper's probabilistic channel model.
- Developed an $\mathcal{O}(n^{1.5})$ algorithm for computing candidate trimmed FFTs. Reported a 1.70–2.49 dB lower SNR threshold than NeLoRa at 90% symbol-decoding accuracy.

Neural and multi-gateway decoding (SRLoRa, MobiHoc 2023; NELoRa-Bench, ICLR Workshop 2023; first author)

- Designed a neural architecture with max-min-merge layers that fuses weak copies of one packet from multiple gateways.
- Released NELoRa-Bench, an open dataset of 27,329 real LoRa symbols for neural demodulation; contributed the dataset and benchmarking to LoRaSeek (MobiCom 2025).

Agricultural IoT (Demeter, MobiCom 2024; Kairos, ICNP 2026)

- Contributed algorithm design and the overall presentation of both papers on cross-soil LoRa communication.

Open-Source Software and Datasets

- LoRaTrimmer decoder: github.com/LoRaTrimmer/LoRaTrimmer, with a reproducible workflow on Kaggle
- NELoRa-Bench dataset: github.com/daibiaoxuwu/NeLoRa_Dataset and Kaggle
- Browser-based interactive LoRa receiver demos: daibiaoxuwu.github.io

Proposal Development

- National Science Foundation: "Physics-Informed Intelligent Signal Processing for Next-Generation Long-Range IoT Communication." Key personnel; PI: Zhichao Cao. New proposal, under review.

Teaching Experience

CSE 410 Operating Systems, Michigan State University, Spring 2026
Undergraduate course; 101 students enrolled. Teaching assistance:

- Redesigned homework as scaffolded operating-system implementation exercises in which students completed missing components.
- Asked students to explain their code and answer questions; provided feedback on efficiency and corner cases and encouraged independent solutions.

Mentoring

- Jingkai Lin, Ph.D. student, Michigan State University. Helped develop the exploration–exploitation reasoning for adaptive transmission control in Kairos (IEEE ICNP 2026; Lin as first author). Also coauthored DeRa (ACM MobiCom 2026; Lin as second author).

Professional Service

- Technical Program Committee member, IEEE International Conference on Mobility, Sensing and Networking (MSN), 2026
- Distinguished Reviewer Board, ACM Transactions on Internet of Things (TIOT), 2026